

☺ LAMP架构部署

前言

LAMP是一个经典的开源技术栈，用于构建和运行动态网站及Web应用。它由以下四个组件组成：

- 1. **Linux**：操作系统，提供稳定、安全的环境，是整个架构的基础。
- 2. **Apache**：Web服务器，负责处理HTTP请求和响应，为用户提供静态或动态内容。
- 3. **MySQL**：关系型数据库管理系统，负责存储和管理网站或应用的数据。
- 4. **PHP/Perl/Python**：脚本语言，用于开发动态内容和后端逻辑，与数据库交互，生成动态网页。

这个架构简单、高效、稳定，是许多中小型Web应用开发的首选解决方案。

集群规划信息

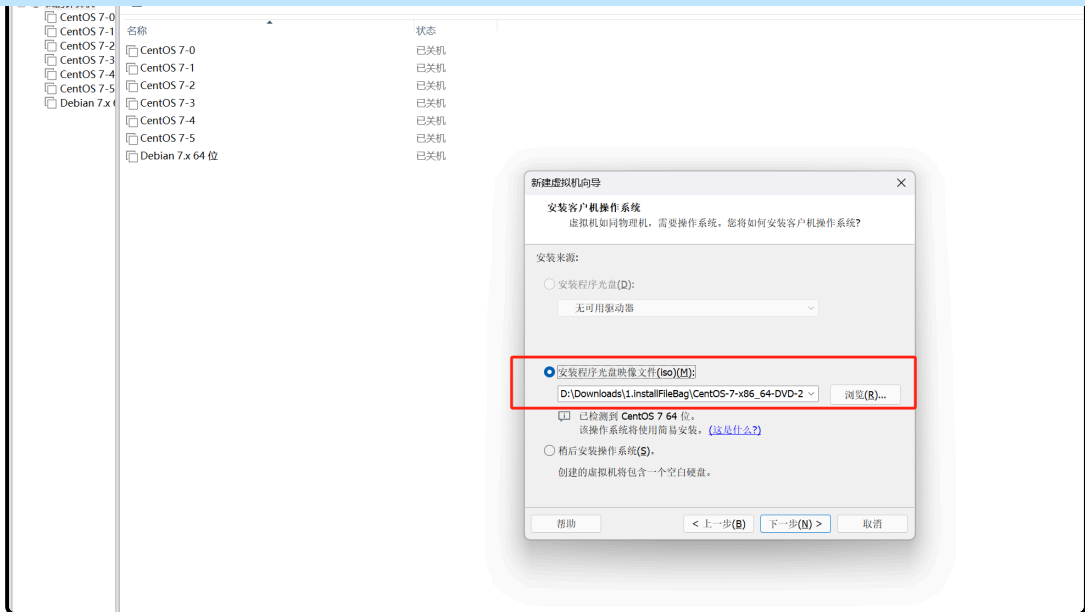
IP:	中间件	版本
192.168.0.103	Apache	Apache/2.4.6 (CentOS)
	MySQL	5.5.68-MariaDB
	PHP	PHP 5.4.16

基础环境部署--操作系统

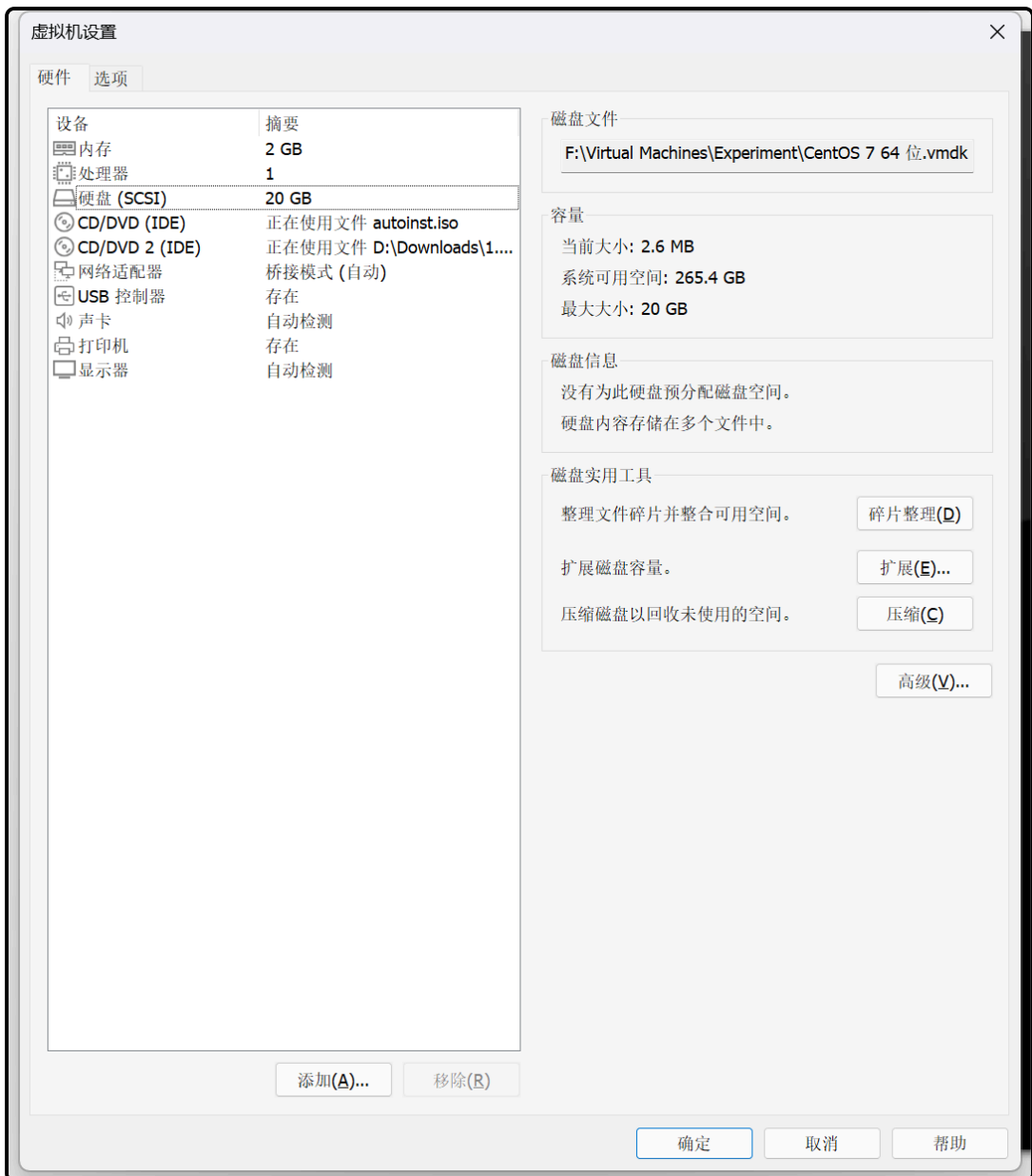
安装包和ISO

 CentOS-7-x86_64-DVD-2009.iso	2023/3/16 7:59	光盘映像文件	4,601,856...
 VMware-workstation-full-17.0.0-20800274.exe	2023/3/16 0:29	应用程序	622,471 KB

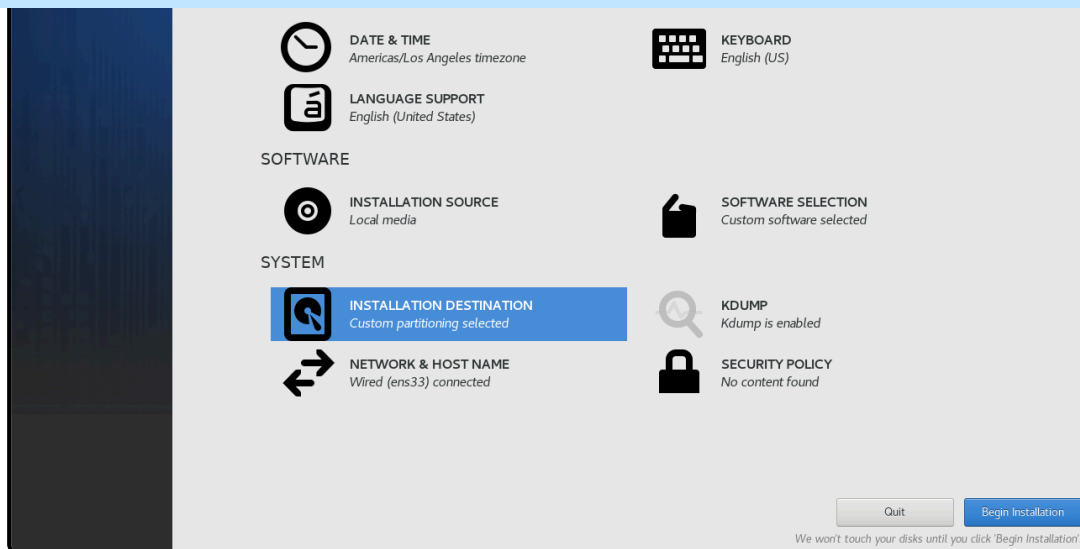
使用VMware Workstation创建虚拟机，选中已有的镜像文件



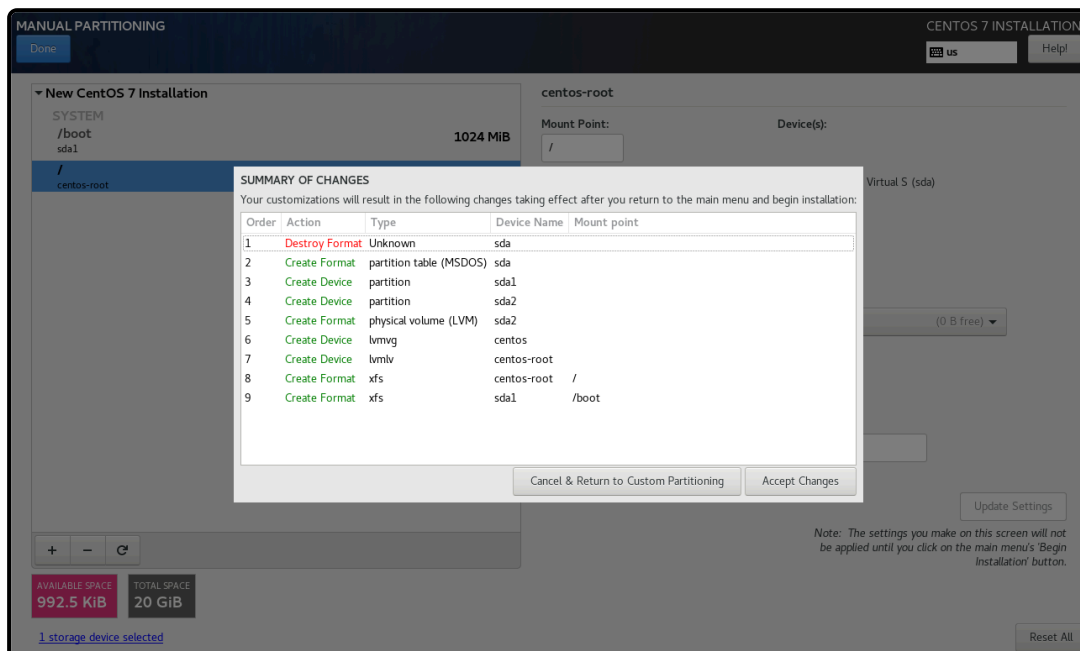
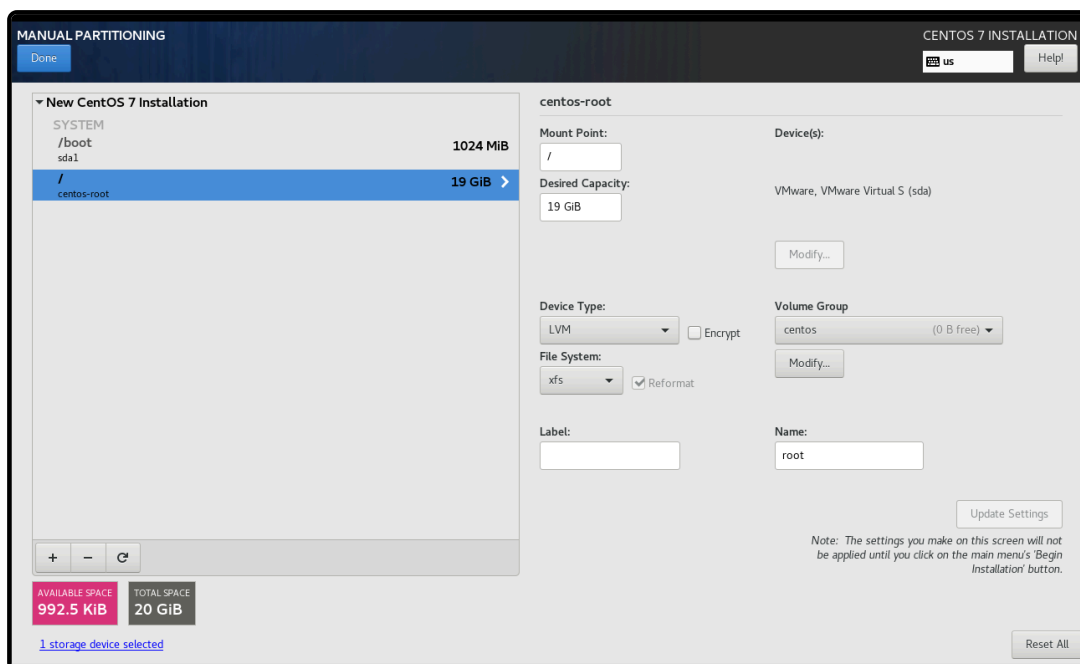
硬件配额设置



自定义文件分区目录



将swap和home都进行删除，保留boot和/分区，将空间大小都分配给 /



```
1 | swapoff -a && sed -ri 's/.*swap.*#&/' /etc/fstab
2 | sed -i 's/^SELINUX=.*SELINUX=disabled/' /etc/selinux/config
```

2) 修改主机名称

```
1 | sudo hostnamectl sethostname leijindong
2 | bash
```

3) 查看虚拟机IP地址

3.1 查看内网IP

```
1 | ip a ls ens33
2
3 | # ens33 为网卡名称
```

3.2 查看公网IP

```
1 | curl http://ip.me
```

3.3 查看bash登陆用户

```
1 | w
```

```
root@leijindong ~]# ip a ls ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:02:0c:59 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.183/24 brd 192.168.0.255 scope global noprefixroute dynamic ens33
        valid_lft 76749sec preferred_lft 76749sec
    inet6 fe80::833:f5c5:db05:eaf3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@leijindong ~]# curl http://ip.me
36.113.134.129
root@leijindong ~]# curl -I http://www.baidu.com
HTTP/1.1 200 OK
Accept-Ranges: bytes
Cache-Control: private, no-cache, no-store, proxy-revalidate, no-transform
Connection: keep-alive
Content-Length: 277
Content-Type: text/html
Date: Mon, 23 Dec 2024 05:16:58 GMT
Etag: "575e1f60-115"
Last-Modified: Mon, 13 Jun 2016 02:50:00 GMT
Pragma: no-cache
Server: bfe/1.0.8.18

root@leijindong ~]# w
21:17:03 up 2:41, 2 users, load average: 0.05, 0.03, 0.05
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT
root     tty1                19:25    7.00s   0.78s   0.03s  w
root@leijindong ~]# _
```

📁 1 安装

1.1) 配置阿里云yum源

```
1 # 备份yum源
2 mv /etc/yum.repos.d/CentOS-Base.repo /etc/yum.repos.d/CentOS-Base.repo.backup
3 # 使用阿里云 yum 源
4 wget -O /etc/yum.repos.d/CentOS-Base.repo https://mirrors.aliyun.com/repo/Centos-7.repo
5 # 生成缓存
6 yum clean all && yum makecache
```

1.2) yum安装Apache:

```
1 yum install httpd -y
2 systemctl enable httpd --now
3
4 [root@leijindong ~]# systemctl status httpd
5 ● httpd.service - The Apache HTTP Server
6    Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset:
disabled)
7    Active: active (running) since Sun 2024-12-22 21:21:18 PST; 6s ago
8    Docs: man:httpd(8)
9          man:apachectl(8)
10   Main PID: 59940 (httpd)
11   Status: "Processing requests..."
12   Tasks: 6
13   CGroup: /system.slice/httpd.service
14           └─59940 /usr/sbin/httpd -DFOREGROUND
15           └─59941 /usr/sbin/httpd -DFOREGROUND
16           └─59942 /usr/sbin/httpd -DFOREGROUND
17           └─59943 /usr/sbin/httpd -DFOREGROUND
18           └─59944 /usr/sbin/httpd -DFOREGROUND
19           └─59945 /usr/sbin/httpd -DFOREGROUND
20
21 Dec 22 21:21:07 leijindong systemd[1]: Starting The Apache HTTP Server...
22 Dec 22 21:21:13 leijindong httpd[59940]: AH00558: httpd: Could not reliably determine
the server's fully qualified domain name, using fe80::833:f5c5:db05:eaf3. Set the
'ServerName' directive globally to suppress this message
23 Dec 22 21:21:18 leijindong systemd[1]: Started The Apache HTTP Server.
```

1.3) 修改防火墙规则

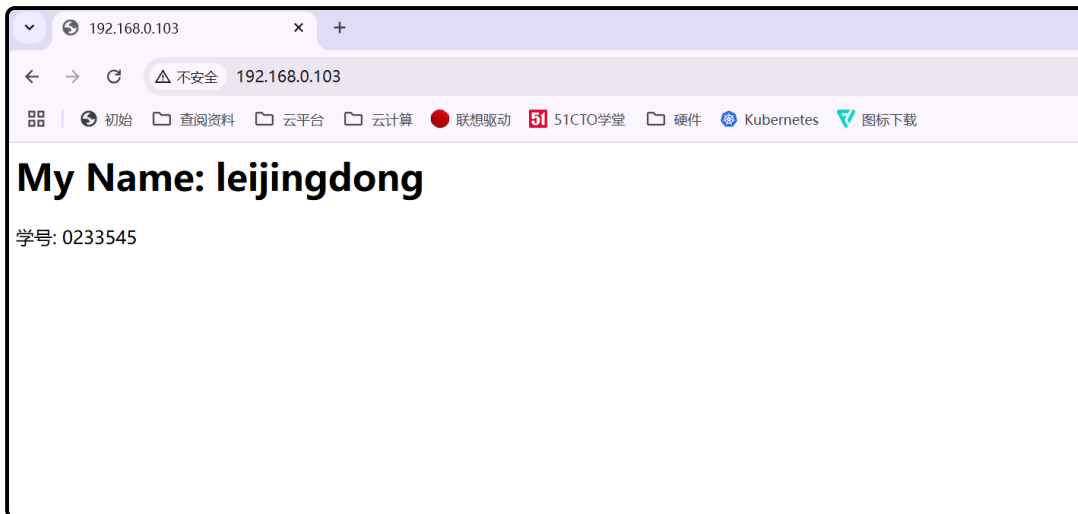
```
1 [root@leijindong ~]# sudo firewall-cmd --permanent --add-service=http
2 success
3 [root@leijindong ~]# sudo firewall-cmd --reload
4 success
```

📁 2 修改访问默认页面进行验证

2.1) 修改权限

```
1 echo "<h1>My Name: leijingdong</h1><p>学号: 0233545</p>" > /var/www/html/index.html
```

```
1 [root@leijindong ~]# curl http://localhost:80
2 <h1>My Name: leijingdong</h1><p>学号: 0233545</p>
3 [root@leijindong ~]# curl -I http://localhost:80
4 HTTP/1.1 200 OK
5 Date: Mon, 23 Dec 2024 05:32:31 GMT
6 Server: Apache/2.4.6 (CentOS)
7 Last-Modified: Mon, 23 Dec 2024 05:22:03 GMT
8 ETag: "34-629e92d43b9dd"
9 Accept-Ranges: bytes
10 Content-Length: 52
11 Content-Type: text/html; charset=UTF-8
```



2. Mysql

📖 2.1 yum安装Mysql:

```
1 yum install mariadb-server -y
2 systemctl enable mariadb --now
3
4 [root@leijindong ~]# systemctl enable mariadb --now
5 Created symlink from /etc/systemd/system/multi-user.target.wants/mariadb.service to
   /usr/lib/systemd/system/mariadb.service.
6 [root@leijindong ~]# systemctl status mariadb
7 ● mariadb.service - MariaDB database server
8    Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; vendor preset:
           disabled)
9    Active: active (running) since Sun 2024-12-22 21:30:46 PST; 2s ago
10   Process: 60246 ExecStartPost=/usr/libexec/mariadb-wait-ready $MAINPID (code=exited,
           status=0/SUCCESS)
11   Process: 60162 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir %n (code=exited,
           status=0/SUCCESS)
12   Main PID: 60245 (mysqld_safe)
13     Tasks: 20
14    CGroup: /system.slice/mariadb.service
15            └─60245 /bin/sh /usr/bin/mysqld_safe --basedir=/usr
16              └─60410 /usr/libexec/mysqld --basedir=/usr --datadir=/var/lib/mysql --plugin-
                 dir=/usr/lib64/mysql/plugin --log-error=/var/log/mariadb/mariadb.log --pid-
                 file=/var/run/mariadb/mariadb.pid --socket=/var/lib/mysql/mysql.sock
17
18 Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: MySQL manual for more
   instructions.
```

```

22 Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: http://dev.mysql.com
23 Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: Consider joining MariaDB's
    strong and vibrant community:
24 Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: https://mariadb.org/get-
    involved/
25 Dec 22 21:30:44 leijindong mysqld_safe[60245]: 241222 21:30:44 mysqld_safe Logging to
    '/var/log/mariadb/mariadb.log'.
26 Dec 22 21:30:44 leijindong mysqld_safe[60245]: 241222 21:30:44 mysqld_safe Starting
    mysqld daemon with databases from /var/lib/mysql
27 Dec 22 21:30:46 leijindong systemd[1]: Started MariaDB database server.

```

```

[Installed:
 mariadb-server.x86_64 1:5.5.68-1.el7

Dependency Installed:
 mariadb.x86_64 1:5.5.68-1.el7      perl-Compress-RawBzip2.x86_64 0:2.061-3.el7      perl-Compress-RawZlib.x86_64 1:2.861-4.el7      perl-DBD-MySQL.x86_64 0:4.023-6.el7      perl-DBD.x86_64 0:1.627-4.el7      perl-Data-Dumper.x86_64 0:2.145-3.el7      perl-IO-Compress.noarch 0:2.061-2.el7
 perl-Net-Daemon.noarch 0:0.48-5.el7      perl-PDPC-mnarch 0:0.2020-14.el7

Installed Packages
 mariadb-server.x86_64 1:5.5.68-1.el7
 mariadb.x86_64 1:5.5.68-1.el7
 perl-Compress-RawBzip2.x86_64 0:2.061-3.el7
 perl-Compress-RawZlib.x86_64 1:2.861-4.el7
 perl-DBD-MySQL.x86_64 0:4.023-6.el7
 perl-DBD.x86_64 0:1.627-4.el7
 perl-Data-Dumper.x86_64 0:2.145-3.el7
 perl-IO-Compress.noarch 0:2.061-2.el7
 perl-Net-Daemon.noarch 0:0.48-5.el7
 perl-PDPC-mnarch 0:0.2020-14.el7

Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; vendor preset: disabled)
Active: active (running) since Sun 2022-12-22 21:30:46 CST; 2s ago
Process: 60240 ExecStartPost=/usr/libexec/mariadb-wait-ready $Mysqld (code=exited, status=0/SUCCESS)
Process: 60240 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir %u (code=exited, status=0/SUCCESS)
Main PID: 60240 (mysqld_safe)
Tasks: 0
CGroup: /system.slice/mariadb.service
         └─60240 /usr/libexec/mysqld --basedir=/usr --datadir=/var/lib/mysql --plugin-dir=/usr/lib64/mysql/plugin --log-error=/var/log/mariadb/mariadb.log --pid-file=/var/run/mariadb/mariadb.pid --socket=/var/lib/mysql/mysql.sock

Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: MySQL Manual for more instructions.
Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: Please report any problems at http://mariadb.org/lira
Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: The latest information about MariaDB is available at http://mariadb.org/.
Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: You can find additional information about the MySQL part at:
Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: http://dev.mysql.com
Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: Consider joining MariaDB's strong and vibrant community:
Dec 22 21:30:43 leijindong mariadb-prepare-db-dir[60162]: https://mariadb.org/get-involved
Dec 22 21:30:44 leijindong mysqld_safe[60245]: 241222 21:30:44 mysqld_safe Logging to '/var/log/mariadb/mariadb.log'.
Dec 22 21:30:44 leijindong mysqld_safe[60245]: 241222 21:30:44 mysqld_safe Starting mysqld daemon with databases from /var/lib/mysql
Dec 22 21:30:46 leijindong systemd[1]: Started MariaDB database server.
]

```

2.2 配置数据库

2.2.1) 安全配置

```

1  # 设置root密码，按提示完成配置。
2  mysql_secure_installation
3
4  #修改端口号为 13306,提高安全性
5  [root@leijindong ~]# cat /etc/my.cnf
6  [mysqld]
7  port=13306
8  datadir=/var/lib/mysql
9  socket=/var/lib/mysql/mysql.sock
10 # Disabling symbolic-links is recommended to prevent assorted security risks
11 symbolic-links=0
12 # Settings user and group are ignored when systemd is used.
13 # If you need to run mysqld under a different user or group,
14 # customize your systemd unit file for mariadb according to the
15 # instructions in http://fedoraproject.org/wiki/Systemd
16
17 [mysqld_safe]
18 log-error=/var/log/mariadb/mariadb.log
19 pid-file=/var/run/mariadb/mariadb.pid
20
21 #
22 # include all files from the config directory
23 #
24 !includedir /etc/my.cnf.d
25
26 [root@leijindong ~]# systemctl restart mariadb
27 [root@leijindong ~]# sudo netstat -tuln | grep 13306
28 tcp        0      0 0.0.0.0:13306      0.0.0.0:*          LISTEN

```

```

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
you haven't set the root password yet, the password will be blank,
so you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB
root user without the proper authorisation.

Set root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n]
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n]
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

```

2.2.2) 登录数据库:

```

1 [root@leijindong ~]# mysql -u root -p
2 Enter password:
3 Welcome to the MariaDB monitor.  Commands end with ; or \g.
4 Your MariaDB connection id is 2
5 Server version: 5.5.68-MariaDB MariaDB Server
6
7 Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
8
9 Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```

2.2.3) 创建数据库和用户, 插入记录:

```

1 CREATE DATABASE testdb;
2 USE testdb;
3 CREATE TABLE testtable (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100));
4 INSERT INTO testtable (name) VALUES ('leijindong');
5 MariaDB [testdb]> SELECT * FROM testtable;
6 +-----+
7 | id | name      |
8 +-----+
9 | 1  | leijindong |
10 +-----+
11 1 row in set (0.00 sec)

```



```

1 [root@leijindong ~]# sudo firewall-cmd --permanent --add-port=13306/tcp
2 success
3 [root@leijindong ~]# sudo firewall-cmd --reload
4 success
5 [root@leijindong ~]# sudo firewall-cmd --list-ports
6 13306/tcp

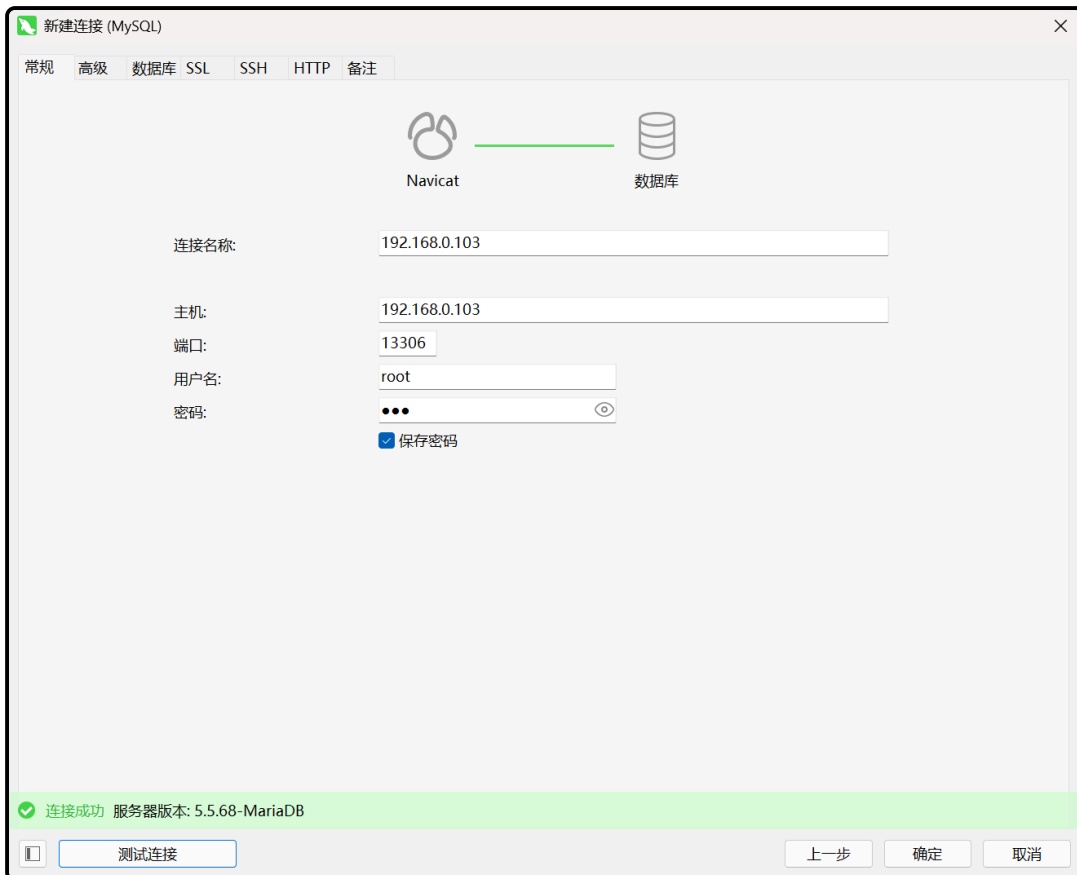
```

2.3.5) 配置远程访问

```

1 #注释掉或删除 bind-address=127.0.0.1
2
3 #授予远程权限:
4 GRANT ALL PRIVILEGES ON *.* TO 'root'@'%' IDENTIFIED BY '123' WITH GRANT OPTION;
5 FLUSH PRIVILEGES;
6
7 # 重启MySQL服务:
8 systemctl restart mariadb

```



3.部署PHP/Perl/Python应用

3.1 安装PHP

```
1 yum install php php-mysql -y
```

3.2 安装phpMyAdmin

```
1 yum install epel-release -y
2 yum install phpmyadmin -y
```

3.3. 配置phpMyAdmin

备份修改配置文件允许远程访问:

```
1 cp /etc/httpd/conf.d/phpMyAdmin.conf{-bak}
2
3 sed -i -e 's/Require ip 127\.0\.0\.1/Require all granted/g' \
4     -e 's/Require ip ::1/Require all granted/g' \
5     -e 's/Deny from All/Allow from All/g' \
6     -e 's/Allow from 127\.0\.0\.1/Allow from All/g' \
7     -e 's/Allow from ::1/Allow from All/g' \
8     /etc/httpd/conf.d/phpMyAdmin.conf
9
10 [root@leijindong ~]# grep -Ev '^s*(#|$)' /etc/httpd/conf.d/phpMyAdmin.conf
11 Alias /phpMyAdmin /usr/share/phpMyAdmin
12 Alias /phpmyadmin /usr/share/phpMyAdmin
13 <Directory /usr/share/phpMyAdmin/>
14     AddDefaultCharset UTF-8
15     <IfModule mod_authz_core.c>
16         <RequireAny>
17             Require all granted
18             Require all granted
19         </RequireAny>
20     </IfModule>
21     <IfModule !mod_authz_core.c>
22         Order Deny,Allow
23         Allow from All
24         Allow from All
25         Allow from All
26     </IfModule>
27 </Directory>
28 <Directory /usr/share/phpMyAdmin/setup/>
29     <IfModule mod_authz_core.c>
30         <RequireAny>
31             Require all granted
32             Require all granted
33         </RequireAny>
34     </IfModule>
35     <IfModule !mod_authz_core.c>
36         Order Deny,Allow
37         Allow from All
38         Allow from All
39         Allow from All
40     </IfModule>
```

```

46 </IfModule !mod_authz_core.c>
47     Order Deny,Allow
48     Allow from All
49     Allow from None
50 </IfModule>
51 </Directory>
52 <Directory /usr/share/phpMyAdmin/setup/lib/>
53     <IfModule mod_authz_core.c>
54         Require all denied
55     </IfModule>
56     <IfModule !mod_authz_core.c>
57         Order Deny,Allow
58         Allow from All
59         Allow from None
60     </IfModule>
61 </Directory>
62 <Directory /usr/share/phpMyAdmin/setup/frames/>
63     <IfModule mod_authz_core.c>
64         Require all denied
65     </IfModule>
66     <IfModule !mod_authz_core.c>
67         Order Deny,Allow
68         Allow from All
69         Allow from None
70     </IfModule>
71 </Directory>

```

📁 3.4 创建PHP应用

3.4.1 创建PHP测试文件:

```
1 echo "<?php phpinfo(); ?>" > /var/www/html/info.php
```

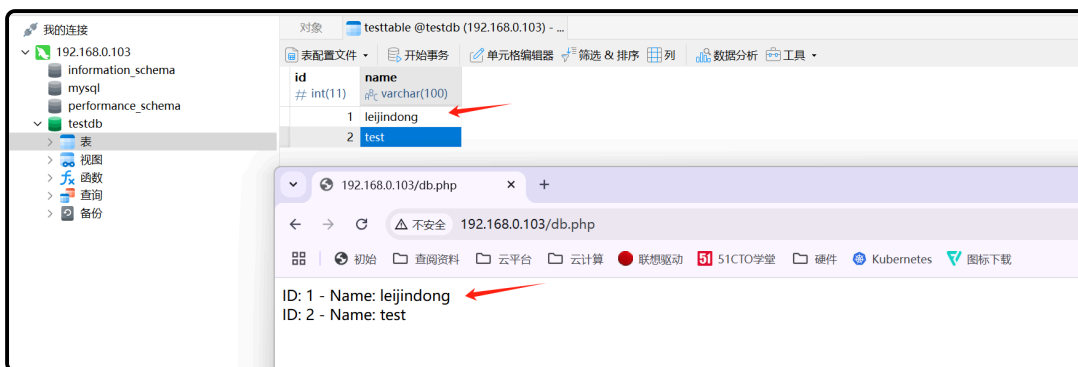
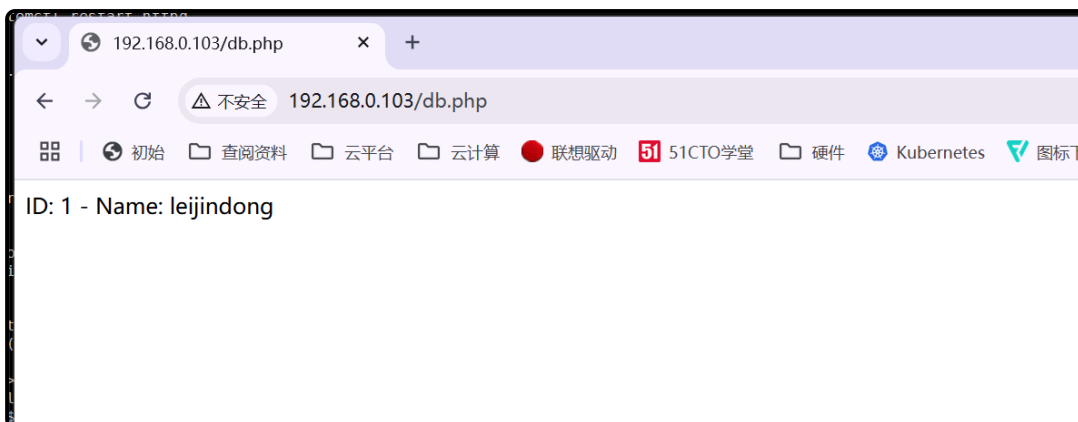
3.4.2 创建连接数据库的PHP脚本 (/var/www/html/db.php):

```

1 cat > /var/www/html/db.php <<'EOF'
2 <?php
3 $servername = "192.168.0.103"; // 数据库服务器地址
4 $username = "root";          // 数据库用户名
5 $password = "123";           // 数据库密码
6 $dbname = "testdb";          // 数据库名称
7 $port = 13306;               // MariaDB 端口号
8
9 // 创建连接
10 $conn = new mysqli($servername, $username, $password, $dbname, $port);
11
12 // 检查连接
13 if ($conn->connect_error) {
14     die("Connection failed: " . $conn->connect_error);
15 }
16
17 $sql = "SELECT * FROM testtable";
18 $result = $conn->query($sql);
19
20 if ($result->num_rows > 0) {
21     while($row = $result->fetch_assoc()) {
22         echo "ID: " . $row["id"]. " - Name: " . $row["name"]. "<br>";
23     }
24 } else {
25     echo "0 results";
26 }

```

```
1 [root@leijindong ~]# curl -I http://127.0.0.1:80/db.php
2 HTTP/1.1 200 OK
3 Date: Mon, 23 Dec 2024 07:10:55 GMT
4 Server: Apache/2.4.6 (CentOS) PHP/5.4.16
5 X-Powered-By: PHP/5.4.16
6 Content-Type: text/html; charset=UTF-8
7
8 [root@leijindong ~]# curl http://127.0.0.1:80/db.php
9 ID: 1 - Name: leijindong<br>
```



4.邮件服务

📁 4.1 安装并配置自启

```
1 yum install mailx sendmail -y
2 systemctl enable sendmail --now
```

📁 4.2 修改配置文件

```
1 cp /etc/mail.rc{,.bak}
```

```
2 set smtp=smtp://smtp.163.com:25
3 set smtp-auth=login
4 set smtp-auth-user=13121423367@163.com
5 set smtp-auth-password=你的网易邮箱授权码
6 set from=13121423367@163.com
7 EOF
```

4.3 邮件测试验证

```
1 echo "测试邮件内容" | mail -s "测试邮件标题" 13121423367@163.com
```

测试邮件标题



13121423367

发至 13121423367 详情

测试邮件内容

2. 编写脚本

```
1 AUTO_MODE=true ./task_manager.sh
```

```
1 #!/bin/bash
2
3 # 设置模式变量: AUTO_MODE=true 则为自动运行模式
4 AUTO_MODE=false # 默认为手动模式
5
6 # 日志文件路径和备份目录
7 APACHE_LOG_DIR="/var/log/httpd" # 实际路径
8 ACCESS_LOG="$APACHE_LOG_DIR/access_log" # 访问日志
9 ERROR_LOG="$APACHE_LOG_DIR/error_log" # 错误日志
10 BACKUP_DIR="/backup/apache_logs"
11 MONITOR_LOG="/var/log/service_monitor.log"
12 RESOURCE_MONITOR_LOG="/var/log/resource_monitor.log"
13
14 # 邮件通知设置
15 ADMIN_EMAIL="13121423367@163.com"
16
17 # 检查和创建备份目录
18 function ensure_backup_dir {
19     if [ ! -d "$BACKUP_DIR" ]; then
20         mkdir -p "$BACKUP_DIR"
21     fi
22 }
23
24 # 任务1: 清理和备份超过30天的日志
25 function backup_and_clean_logs {
```

```

30 }
31
32 # 任务2: 实时监控服务运行状态
33 function monitor_services {
34     echo "开始监控服务状态..."
35     SERVICES=("httpd" "mariadb") # 可根据需要调整服务名称
36     for service in "${SERVICES[@]"; do
37         if ! systemctl is-active --quiet "$service"; then
38             echo "${date): 服务 $service 已停止, 正在尝试重启..." | tee -a "$MONITOR_LOG"
39             systemctl restart "$service"
40             if ! systemctl is-active --quiet "$service"; then
41                 echo "${date): 服务 $service 重启失败, 通知管理员。" | tee -a
"$MONITOR_LOG"
42                 echo "服务 $service 重启失败, 请检查服务器。" | mail -s "服务故障通知"
"$ADMIN_EMAIL"
43             else
44                 echo "${date): 服务 $service 重启成功。" | tee -a "$MONITOR_LOG"
45             fi
46         else
47             echo "${date): 服务 $service 正常运行。" | tee -a "$MONITOR_LOG"
48         fi
49     done
50 }
51
52 # 任务3: Apache 日志流量分析 (可选功能)
53 function analyze_logs {
54     echo "开始分析 Apache 日志..."
55     if [ -f "$ACCESS_LOG" ]; then
56         echo "访问最多的 IP 地址: "
57         awk '{print $1}' "$ACCESS_LOG" | sort | uniq -c | sort -nr | head -10
58
59         echo "访问最多的 URL: "
60         awk '{print $7}' "$ACCESS_LOG" | sort | uniq -c | sort -nr | head -10
61
62         echo "返回码统计: "
63         awk '{print $9}' "$ACCESS_LOG" | sort | uniq -c | sort -nr
64     else
65         echo "无法找到 Apache 访问日志! "
66     fi
67 }
68
69 # 任务4: 实时监控资源使用情况 (修改后实现实时监控并只显示最新结果)
70 function monitor_resources {
71     echo "开始实时监控资源使用情况..."
72     > "$RESOURCE_MONITOR_LOG" # 每次清空文件
73
74     while true; do
75         # 写入最新监控结果
76         echo "=====${date '+%Y-%m-%d %H:%M:%S'}=====" > "$RESOURCE_MONITOR_LOG"
77         echo "CPU 和内存使用情况: " >> "$RESOURCE_MONITOR_LOG"
78         top -b -n 1 | head -n 10 >> "$RESOURCE_MONITOR_LOG"
79
80         echo "" >> "$RESOURCE_MONITOR_LOG"
81         echo "磁盘 I/O 情况: " >> "$RESOURCE_MONITOR_LOG"
82         iostat -x 1 1 | tail -n +3 >> "$RESOURCE_MONITOR_LOG"
83
84         echo "" >> "$RESOURCE_MONITOR_LOG"
85         echo "网络流量: " >> "$RESOURCE_MONITOR_LOG"
86         ifstat 1 1 | tail -n 1 >> "$RESOURCE_MONITOR_LOG"
87
88         # 清屏并打印最新监控结果
89         clear
90         cat "$RESOURCE_MONITOR_LOG"
91
92         # 间隔时间 (例如 5 秒)
93         sleep 5
94     done
95 }

```

```

102     monitor_services
103     exit 0
104 fi
105
106 # 手动模式菜单
107 while true; do
108     echo "请选择要执行的任务: "
109     echo "1) Apache 日志流量分析"
110     echo "2) 清理和备份 Apache 日志"
111     echo "3) 实时监控资源使用情况"
112     echo "4) 实时监控服务运行状态"
113     echo "5) 全部执行"
114     echo "6) 退出"
115     read -p "输入选项 (1-6): " choice
116
117     case $choice in
118         1) analyze_logs ;;
119         2) backup_and_clean_logs ;;
120         3) monitor_resources ;; # 实时监控资源
121         4) monitor_services ;;
122         5)
123             analyze_logs
124             backup_and_clean_logs
125             monitor_resources &
126             monitor_services
127             echo "所有任务执行完成! "
128             ;;
129         6) exit 0 ;;
130         *) echo "无效选项, 请重试! " ;;
131     esac
132 done
133 }
134
135 # 启动脚本主程序
136 main_menu
137

```

3.脚本验证

3.1 Apache 日志流量分析

```

[root@leijindong ~]# bash task_manager.sh
请选择要执行的任务:
1) Apache 日志流量分析
2) 清理和备份 Apache 日志
3) 实时监控资源使用情况
4) 实时监控服务运行状态
5) 全部执行
6) 退出
输入选项 (1-6): 1
开始分析 Apache 日志...
访问最多的 IP 地址:
  12 192.168.0.100
   4 127.0.0.1
   4 ::1
访问最多的 URL:
   7 /db.php
   7 /
   5 408
   1 /favicon.ico
返回码统计:
  13 200
   5 "-"
   1 404
   1 304
请选择要执行的任务:
1) Apache 日志流量分析
2) 清理和备份 Apache 日志
3) 实时监控资源使用情况
4) 实时监控服务运行状态
5) 全部执行
6) 退出
输入选项 (1-6): 6
[root@leijindong ~]#

```

请选择要执行的任务：
1) Apache 日志流量分析
2) 清理和备份 Apache 日志
3) 实时监控资源使用情况
4) 实时监控服务运行状态
5) 全部执行
6) 退出
输入选项 (1-6): 2
开始备份和清理日志...
日志备份完成，保存在 /backup/apache_logs。
请选择要执行的任务：
1) Apache 日志流量分析
2) 清理和备份 Apache 日志
3) 实时监控资源使用情况
4) 实时监控服务运行状态
5) 全部执行
6) 退出
输入选项 (1-6):

3.实时监控资源使用情况

```
[root@leijindong ~]# bash task_manager.sh
请选择要执行的任务:
1) Apache 日志流量分析
2) 清理和备份 Apache 日志
3) 实时监控资源使用情况
4) 实时监控服务运行状态
5) 全部执行
6) 退出
输入选项 (1-6): 3
开始实时监控资源使用情况...
===== 2024-12-23 00:51:40 =====
CPU 和内存使用情况:
top - 00:51:40 up 6:16, 4 users, load average: 0.00, 0.01, 0.05
Tasks: 137 total, 1 running, 136 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 16.7 sy, 0.0 ni, 83.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 1863004 total, 234768 free, 485720 used, 1142516 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 1186056 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 128524    7204  4204 S   0.0   0.4   0:00.78 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.02 kthreadd
    4 root        0  -20      0      0      0 S   0.0   0.0   0:00.00 kworker/0:0H

磁盘 I/O 情况:
avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           1.12    0.05   0.91   0.03    0.00   97.89

Device:            rrqm/s   wrqm/s     r/s     w/s    kB/s    kB/s avgrq-sz avgrq-sz  await  r_await w_await  svctm  %util
sda               0.00     0.07    1.73    0.74   52.45   44.45    78.47     0.00    0.83    0.68    1.19    0.42    0.10
scd0              0.00     0.00    0.00    0.00    0.00    0.00     8.00     0.00    2.25    2.25    0.00    2.25    0.00
scd1              0.00     0.00    0.00    0.00    0.00    0.00    114.22    0.00    3.61    3.61    0.00    3.06    0.00
dm-0              0.00     0.00    1.65    0.81   51.72   43.75    77.77     0.00    0.92    0.71    1.33    0.42    0.10

网络流量:
===== 2024-12-23 00:51:45 =====
CPU 和内存使用情况:
top - 00:51:45 up 6:16, 4 users, load average: 0.00, 0.01, 0.05
Tasks: 137 total, 1 running, 136 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 6.2 sy, 0.0 ni, 93.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 1863004 total, 234772 free, 485584 used, 1142648 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 1186192 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
64712 root        20   0 161968   2124   1512 R   6.7   0.1   0:00.01 top
    1 root        20   0 128524    7204  4204 S   0.0   0.4   0:00.79 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.02 kthreadd

磁盘 I/O 情况:
avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           1.12    0.05   0.91   0.03    0.00   97.89

Device:            rrqm/s   wrqm/s     r/s     w/s    kB/s    kB/s avgrq-sz avgrq-sz  await  r_await w_await  svctm  %util
sda               0.00     0.07    1.73    0.74   52.44   44.44    78.47     0.00    0.83    0.68    1.19    0.42    0.10
scd0              0.00     0.00    0.00    0.00    0.00    0.00     8.00     0.00    2.25    2.25    0.00    2.25    0.00
scd1              0.00     0.00    0.00    0.00    0.00    0.00    114.22    0.00    3.61    3.61    0.00    3.06    0.00
dm-0              0.00     0.00    1.65    0.81   51.71   43.75    77.77     0.00    0.92    0.71    1.33    0.42    0.10
```

3.4 实时监控服务运行状态

开始监控资源使用情况...
请选择要执行的任务：
1) Apache 日志流量分析
2) 清理和备份 Apache 日志
3) 实时监控资源使用情况
4) 实时监控服务运行状态
5) 全部执行
6) 退出
输入选项 (1-6): 4
开始监控服务状态...
Mon Dec 23 00:44:22 PST 2024: 服务 httpd 正常运行。
Mon Dec 23 00:44:22 PST 2024: 服务 mariadb 正常运行。

服务故障通知



13121423367

发至 13121423367 详情

🔍 备注 ...

服务 httpd 重启失败，请检查服务器。

📖 1.设置crontab定时任务

1.1 编写crontab 规则

```
1 crontab -e
2 # 每五分钟执行一次
3 */5 * * * * /root/task_manager.sh >> /var/log/server_monitor_cron.log 2>&1
```

```
1 [root@leijindong ~]# crontab -l
2 */5 * * * * /root/task_manager.sh >> /var/log/server_monitor_cron.log 2>&1
3
```

📖 2.设置firewalld规则

2.1 禁止被ping

```
1 firewall-cmd --permanent --add-icmp-block=echo-request
2
3 #重新加载防火墙规则
4 firewall-cmd --reload
```